

PCB DESIGN ASSIGNMENT

ESP32 + LoRa Sensor Interface Board

Applicant: Anshul Sharma

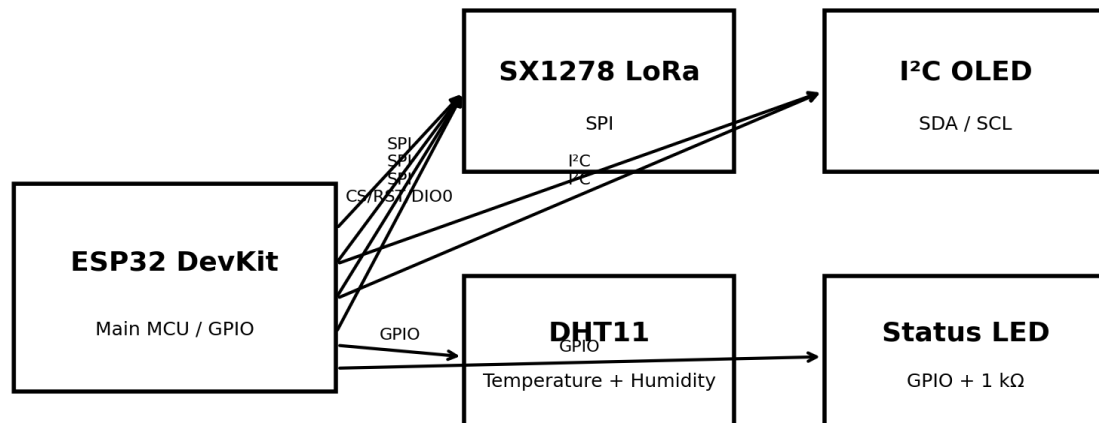
Target internship: Embedded Systems / PCB Design

Design tool: KiCad (basic-level knowledge)

Submission note: This assignment is presented as a PCB design work sample based on my existing academic exposure to ESP32, LoRa and Arduino. I am at the beginner stage in PCB design and have not represented this as a manufactured or production-tested PCB.

1. Objective

The objective is to develop a compact embedded-system PCB concept around an ESP32 controller. The board provides interfaces for an SX1278 LoRa module, DHT11 temperature/humidity sensor, I²C OLED display and a simple status LED. The design demonstrates the basic PCB-design workflow of circuit planning, interface selection, component placement, routing and design-rule awareness.



ESP32 + LoRa Sensor Interface — Conceptual Schematic

Figure 1. Conceptual system schematic.

2. Circuit / Interface Description

Block	Interface	Purpose
ESP32 ↔ LoRa	SPI	High-speed digital communication with the LoRa module
LoRa control	CS, RST, DIO0	Chip selection, reset and interrupt/status signalling
ESP32 ↔ OLED	I ² C	Display data and control
DHT11	GPIO	Temperature and humidity data
Status LED	GPIO + 1 kΩ	Visual indication while limiting LED current
Power	5 V input / 3.3 V logic	Supply arrangement for the prototype

3. PCB Placement Strategy

The ESP32 is placed near the centre as the main controller. The LoRa module is separated from the other interfaces to keep its antenna/module area accessible. The OLED and DHT11 connectors are placed toward the board edge for convenient external wiring. Power entry is kept away from sensitive signal routing. A continuous ground area is considered on the bottom copper layer.

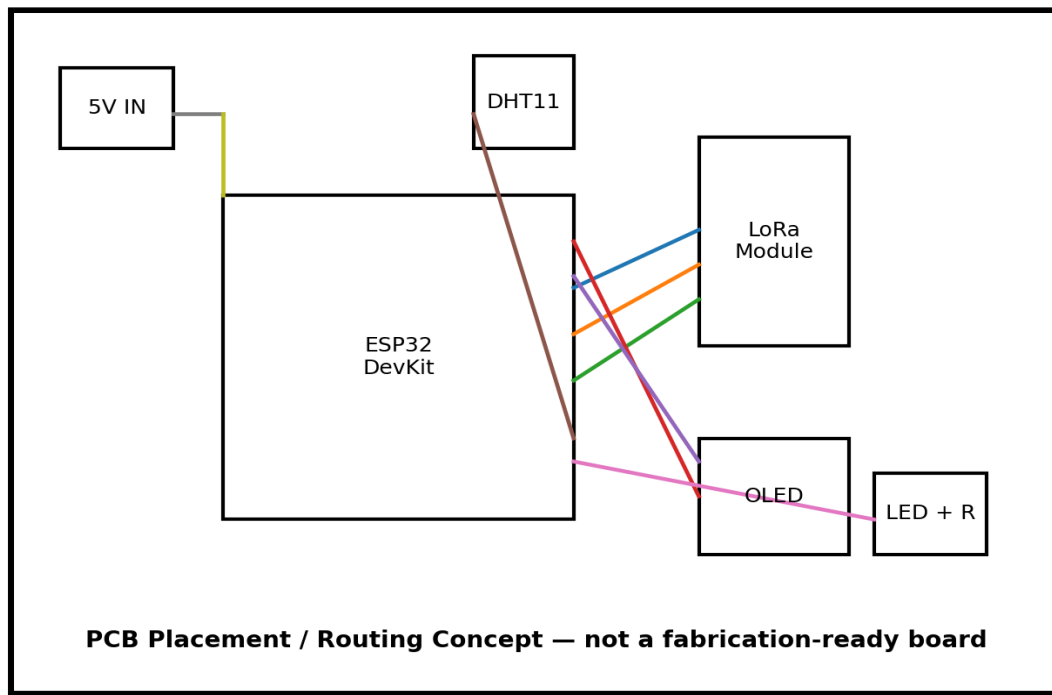


Figure 2. PCB placement and routing concept.

4. PCB Design Considerations

- Use an appropriate board outline on the Edge.Cuts layer.
- Select footprints according to the actual ESP32/LoRa modules and connectors being used.
- Keep power and ground connections short and robust.
- Avoid unnecessary sharp routing and maintain suitable clearance between copper features.
- Provide adequate decoupling near powered modules in a final schematic.
- Keep the LoRa antenna region free from copper/components according to the module manufacturer's recommendations.
- Run ERC on the schematic and DRC on the PCB before manufacturing.

5. KiCad Workflow

The normal KiCad workflow is schematic capture → assign footprints → transfer/update the PCB → component placement → routing → ground/power review → ERC/DRC → manufacturing outputs. KiCad's official documentation describes this schematic-to-PCB workflow and provides ERC/DRC tools for checking electrical and layout-rule problems. ■cite■turn0search13■turn0search12■

6. My Current Skill Level

I have basic knowledge of PCB designing using KiCad and understand the fundamentals of schematic planning, footprints, component placement, tracks, layers and basic routing. My stronger practical academic exposure is in embedded electronics, particularly Arduino, ESP32 and LoRa-based projects. I am applying for this internship specifically to gain hands-on experience in professional PCB design, review, debugging and manufacturing practices.

7. Expected Interview Explanation

Why ESP32? It provides a capable microcontroller platform with GPIO, SPI and I²C interfaces.

Why SPI for LoRa? The SX1278 family commonly uses an SPI host interface for communication with the MCU.

Why I²C for OLED? It reduces the number of signal lines required for the display interface.

Why 1 k Ω with the LED? It limits current through the LED.

What is DRC? Design Rule Check identifies layout issues such as clearance and routing-rule violations.

What would I do before fabrication? Verify the exact module footprints/pinout, complete ERC/DRC, check power requirements and antenna keep-out, then generate manufacturing outputs.

8. Honesty / Scope

This is a learning-level design assignment and concept work sample. It is not claimed to be a manufacturing-ready PCB or a physically fabricated/tested board. The purpose is to demonstrate my understanding of embedded hardware interfaces and my current foundation in PCB design while showing my willingness to learn industry practices.